

Winner Playbook — NQ Regime Volatility Size-Ramp

Instrument: Nasdaq-100 futures (NQ) · Book: 1h + 4h fusion (combined) · Kind: *sizing overlay* (not an entry strategy) · Date: 2026-07-18 · Author: Agent B (research)

🚨 VERDICT: EXPERIMENTAL CANDIDATE — not a confirmed edge

The **signal is validated**: it beats **96%** of random regime → size assignments, helps **4 of 5** purged cross-validation folds and **all 3** calendar years, and its mechanism is understood and independently corroborated on a second layer. But the **dollar magnitude is *not* statistically confirmed**: a block-bootstrap of the uplift gives a 90% confidence interval of **[−21, 075, +61,937]**, which **includes zero** (P(positive) = 70%). **Shipped OFF by default**. **Do not change deployed defaults on it**.

1. The headline

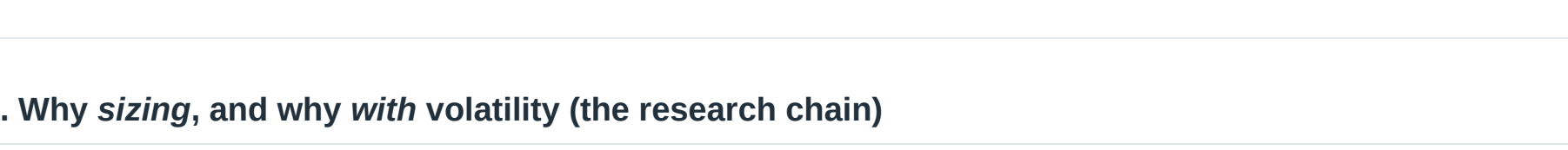
	Profit	max Drawdown	Return/DD	Trades
flat book (baseline)	\$151,872	\$27,508	5.52	539
+ regime size-ramp (equal risk)	\$162,228	\$27,508 (held)	5.90	539
Δ	+\$10,356 (+6.8%)	0	+0.38	0

"Equal risk" is the **honest framing**. Raw, the ramp lifts profit to \$182,927 but also lifts drawdown to \$31,018 — more return for more risk, which proves nothing. So we ****normalize the whole book**** by 'flat_maxDD / ramped_maxDD' (×0.887), pinning max-drawdown to the flat book's ****\$27,508****. The **+\$10,356 is therefore *extra profit at identical risk***, with the **same trades** — none added, none removed.



2. What it actually does (the exact rule)

- For each trade, look up the **day's volatility regime** in a static, precomputed artifact (regime/nq_daily_regime.csv ; 4,977 days; 0 = calmest ... 3 = most turbulent).
- Multiply that trade's P/L by a **linear ramp**: calmest **0.5×** → most turbulent **1.5×**.
- Normalize** the book by flat_maxDD / ramped_maxDD so max-drawdown returns to the flat book's.



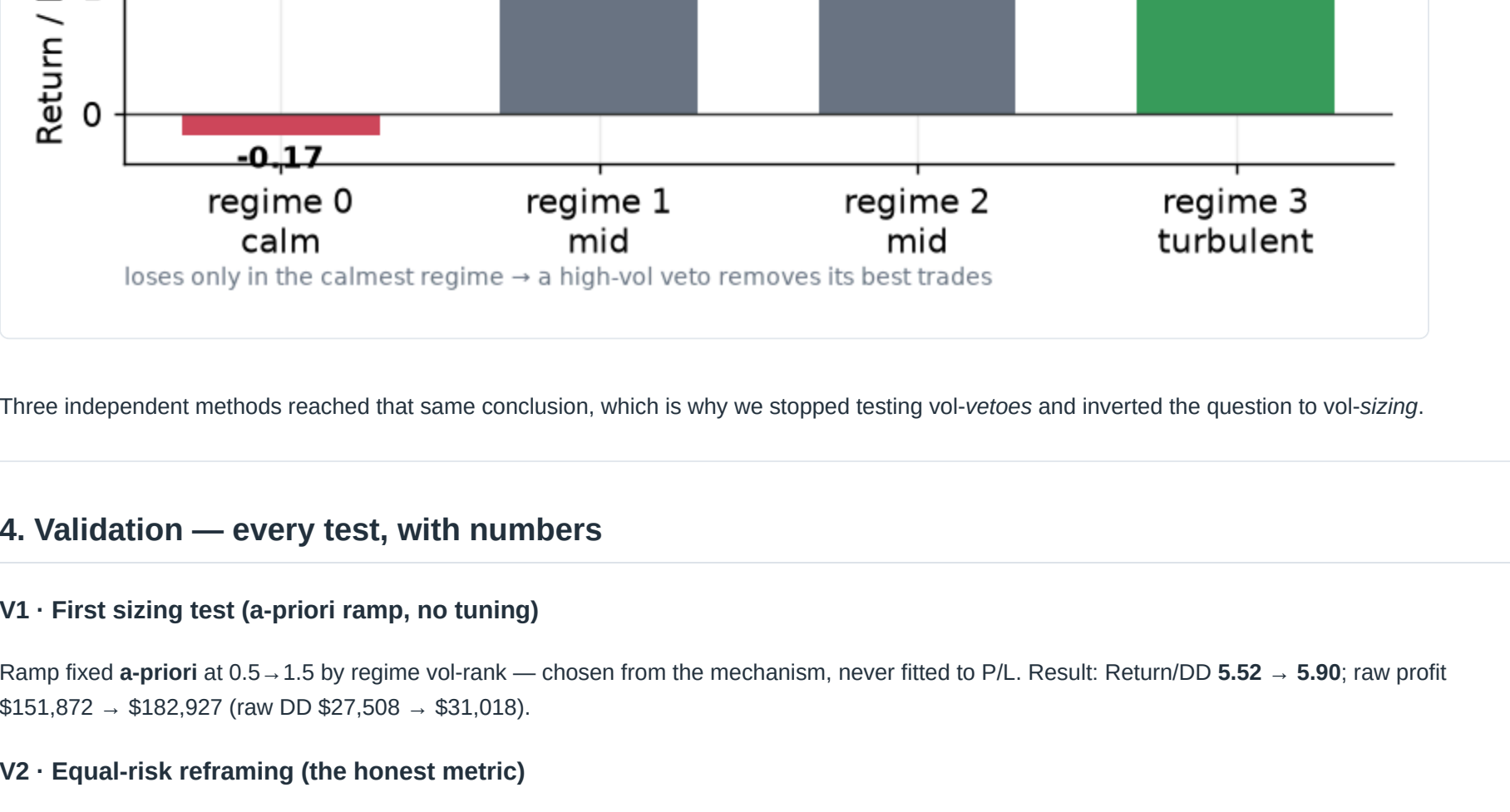
The **regime model**. Gaussian Hidden Markov Model, **4 states**, full covariance, fit on **daily** NQ features — log-return, log intraday realized-volatility, 20-day volume z-score. Parameters are fit on **pre-2024 data only**. Each day's state is the **filtered** (causal forward-algorithm) argmax — *never* the smoothed posterior or the Viterbi path, both of which see the whole sequence and would be look-ahead bias. States are then ranked by realized-volatility emission mean. **Nothing in the pipeline sees the future**.

3. Why sizing, and why with volatility (the research chain)

This overlay is the constructive end of a four-branch investigation whose first three results were all negative — and those negatives are *why* this one is shaped the way it is.

Branch	Attempt	Result
research-timesfm-fusion	skip trades when a foundation model is uncertain	✗ real in-sample, dies out-of-sample
research-regime-hmm	skip trades in the turbulent regime	✗ hurts — and revealed <i>why</i>
research-chronos2	same, with the strongest successor model	✗ identical failure (bands correlate 0.71)
research-regime-edge	size by regime instead of skipping	✓ this

The **discovery that redirected everything**: conditioned on the causal regime, the strategy earns its **best** risk-adjusted return in the **most turbulent** regime and **loses only in the calmest**. It is **vol-seeking**. Every "avoid volatility" rule was therefore removing its best trades.



Three independent methods reached that same conclusion, which is why we stopped testing vol-vetoes and inverted the question to vol-sizing.

4. Validation — every test, with numbers

V1 · First sizing test (a-priori ramp, no tuning)

Ramp fixed **a-priori** at 0.5 → 1.5 by regime vol-rank — chosen from the mechanism, never fitted to P/L. Result: Return/DD **5.52 → 5.90**; raw profit \$151,872 → \$182,927 (raw DD \$27,508 → \$31,018).

V2 · Equal-risk reframing (the honest metric)

Normalizing to hold max-DD at 27,508 : *profit* * \$151,872 → 162,228 **, *i.e.* * +\$10,356 (+6.8%) at identical risk**. This is the number quoted everywhere in this playbook.

V3 · Scale robustness (is 0.5 → 1.5 cherry-picked?)

ramp	Return/DD	equal-risk profit	Δ
0.7 → 1.3	5.76	\$158,380	+\$6,508
0.5 → 1.5	5.90	\$162,228	+\$10,356
0.3 → 1.7	5.89	\$162,144	+\$10,272

Every steepness helps and the benefit **saturates** past 0.5 → 1.5 — so it is a sensible default, **not a tuned peak**. (Picking the *best* scale on this book would be overfitting; we didn't.)

V4 · Out-of-sample holdout

Ramp fixed a-priori, 2026 held out: in-sample (2024–25) **3.79 → 4.19**; held-out **2026 10.76 → 10.85**. Helps on both sides.

V5 · Random-regime control (does the ordering carry information?)

Shuffle which regime gets which multiplier, keeping the same multipliers, 2,000×. The real assignment scores **5.90 vs a median 5.16**, beating **96%** of random maps. → **the regime ordering is informative**, not noise.

V6 · Per-year

2024 **1.15 → 1.37** · 2025 **7.24 → 9.35** · 2026 **10.76 → 10.85**. Helps in 3/3 years.

V7 · Dumb control — the textbook alternative *hurts*

Classic **inverse-volatility targeting** (size ∝ 1/vol), the standard practitioner move, scores Return/DD **4.06 — far below the 5.52 baseline**. It shrinks exactly the turbulent trades this strategy earns on. This is the single cleanest confirmation that the *direction* of the ramp is the point.

V8 · Independent corroboration on a second layer

Applied to the **L2 (secondary 4h) layer alone**: Return/DD **1.81 → 2.13**. A second, independent book agrees. 🚨 But on **L1 (primary 1h) alone it hurts** (6.94 → 6.21), because L1's best regime is *mid*, not the most turbulent. → **the ramp is validated on the combined book, and is NOT uniform per-layer**.

V9 · Block-bootstrap of the uplift — 🚨 the qualifier

Resampling 20-trade blocks 3,000×: median uplift **+11,699 **, but * *\$921,075, +\$60,937]** and **P(uptift > 0) = 70%**. The **dollar magnitude is inside the noise**. With −539 trades over one year, the sampling error is larger than the effect. *This is why the verdict is "candidate", not "confirmed"*.

V10 · Purged 5-fold cross-validation (time-contiguous)

fold	period	flat → ramp
1	2024-01 ... 06	0.24 → 0.26 (+0.02)
2	2024-06 ... 11	0.26 → 0.35 (+0.09)
3	2024-11 ... 2025-05	1.29 → 1.74 (+0.44)
4	2025-05 ... 10	5.28 → 6.67 (+1.39)
5	2025-10 ... 2026-05	9.95 → 9.89 (−0.05)

Helps in **4/5** held-out folds — directionally consistent, magnitude variable.

Validation scoreboard

test	result
V1 a-priori ramp	✓ 5.52 → 5.90
V2 equal-risk	✓ +\$10,356 at identical DD
V3 scale robustness	✓ helps at every steepness
V4 OOS holdout	✓ helps in-sample and 2026
V5 random control	✓ beats 96%
V6 per-year	✓ 3/3
V7 dumb control	✓ textbook inverse-vol hurts (4.06)
V8 second layer	✓ L2 corroborates · 🚨 L1-standalone hurts
V9 bootstrap CI	🚨 includes zero — magnitude unconfirmed
V10 purged CV	✓ 4/5 folds

5. Integration testing — every detail

Design principle: additive-only, fail-closed

The dashboard integration was built so it is **incapable** of altering an existing number:

- optimize/regime_overlay.py returns a **separate dict**; it never mutates meta_boxes / meta.summary .
- The server attaches it **only** when body.regime_sizing is truthy, inside a try/except that swallows everything — any failure leaves the response exactly as it was.
- The module itself returns **None** on any internal problem (missing artifact, <20 trades, no date matches).
- The frontend checkbox defaults **unchecked**, so the default request is byte-identical to before.

I1 · Golden-safety proof (flag on vs off)

Live API call to /api/backtest_causal with the same NQ 1h preset, **flag off** then **on**:

	boxes P/L	boxes max_dd
flag OFF	78,823.21357885841	8,815.127587500028
flag ON	78,823.21357885841	8,815.127587500028

Core numbers byte-identical. meta.regime_sizing present when ON, absent when OFF: ✓ (The repo's perf/check_golden.py could not run — it fails at champion_preset() with a missing ws1m_* preset id. Verified this failure is **pre-existing**: it reproduces identically at the pre-change commit 7a4bd4f . So the API on/off equality above is the golden evidence.)

I2 · 🐛 Bug caught #1 — the overlay silently did nothing

First live run: the card never rendered. Cause: log rows carry time as a **Unix epoch integer** (1735758000), not a date string, so str(time)[:10] produced "1735758000" and matched **zero** regime dates → the function returned **None** every time. **Had this shipped, the feature would have appeared installed and simply never worked**. Fixed with an explicit _day() (epoch → UTC date, string passthrough).

I3 · 🐛 Bug caught #2 — a loss rendered as a gain

The card computed sign = d>=0 ? '+' : '-' — so a **negative delta** printed as **\$18,157** instead of **−\$18,157**. **This is the dangerous class of bug: a plausible-looking wrong number**. Fixed to d>=0 ? '+' : '-' , re-deployed, re-verified in the browser: now reads **−\$18,157** in red.

I4 · Layout defect

The card was rendered into a single narrow grid cell, truncating every value. Fixed with grid-column:1/-1 + explicit ink colour on the amber surface; re-screenshotted to confirm legibility.

I5 · Browser UI wriggle (not just the API)

Driven with playwright against the **live** service, asserting the string end-to-end: select NQ → select 1h → check #l1_regime_sizing (asserted is_checked == True) → click **live** wait → assert the behaviour "Regime size-ramp" appears in #cards → capture the card element and the full page. Card text captured verbatim, including the warning line and Δ profit at equal risk −\$18,157 .

I6 · Deployment chain (deployed == committed)

commit → push → merge to dev (−no-ff, no conflicts) → server git pull --ff-only → dash.sh refresh .Post-deploy dash.sh status : **RUNNING, port 8200, health check HTTP 200**. Server dev tip matches origin.

I7 · Bundle reproduction test

verify.py recomputes every documented figure from the bundled fixture and fails loudly on any drift:

PASS trades	got	539	expected	539
PASS flat P/L	got	151,872.00	expected	151,872.00
PASS flat maxDD	got	27,508.00	expected	27,508.00
PASS flat Return/DD	got	5.52	expected	5.52
PASS overlay P/L	got	162,228.41	expected	162,228.00
PASS overlay maxDD	got	27,508.00	expected	27,508.00
PASS overlay Return/DD	got	5.90	expected	5.90
PASS Δ at equal risk	got	10,356.41	expected	10,356.00
PASS equal-risk scale	got	0.89	expected	0.89
RESULT:	✓ ALL MATCH — reproduces to the dollar			

6. The real dashboard run

Live service, NQ 1h, experimental checkbox **enabled**, Run pressed — the overlay card as it actually renders:



Read this carefully — it is showing a **LOSS**, and that is **correct** behaviour. On the dashboard's default NQ 1h L1 book (850 trades, itself losing −10,683), *thereoverlymakest* * *worse : −18,157*. That book is **not** the book the ramp was validated on (the combined 1h+4h fusion, 539 trades). This is exactly the **"not uniform per-layer"** caveat from V8, visible live — and the card is doing its job as an honest per-book diagnostic rather than flattering the feature.

7. 🚨 When NOT to test this

- Don't run it on other instruments. The regime artifact is **NQ-derived**. Re-derive per instrument.
- Don't apply it per-layer. Validated on the **combined** book. Helps L2 alone; **hurts L1** alone.
- Don't quote the **+\$10,356 as a promise**. Its 90% CI includes zero. Quote the *direction*, not the dollar.
- Don't skip the **equal-risk normalization**. Without it the ramp raises absolute drawdown to \$31,018.
- Don't enable it by default or let it change any champion's deployed numbers. It is opt-in, additive.

8. What would upgrade it to CONFIRMED

A **longer, bear-inclusive trade book** — which requires the **2010–2023 box levels** we do not currently have (they are a scraped external feed, absent pre-2024). With those, the bootstrap either tightens above zero (confirmed) or it doesn't (killed). That single data gap is the whole difference between "candidate" and "edge".

9. Reproduce it yourself

```
pip install -r requirements.txt
python3 verify.py # 9/9 PASS, to the dollar
python3 apply_regime_sizing.py reference/nq_2426_fusion_entries.csv regime/nq_daily_regime.csv --enable
```

10. Provenance

Branch research-regime-edge → merged to dev . Evidence chain: subprojects/regime-edge/docs/ — EXP2_SIZING.md (V1), EXP2b_SIZING_PROMOTED.md (V2–V6), SECOND_TEST.md (V9–V10), EXP3b_L2_BOOK.md (V8), EXP1b_CONCENTRATION_CLEAN.md (a rejected sibling), DEPLOY.md ; cross-bran write-up in reports/MEGA_REPORT.md . Dashboard integration: Parametric-Indicators/optimize/regime_overlay.py , server.py , frontend/dashboard.html .